

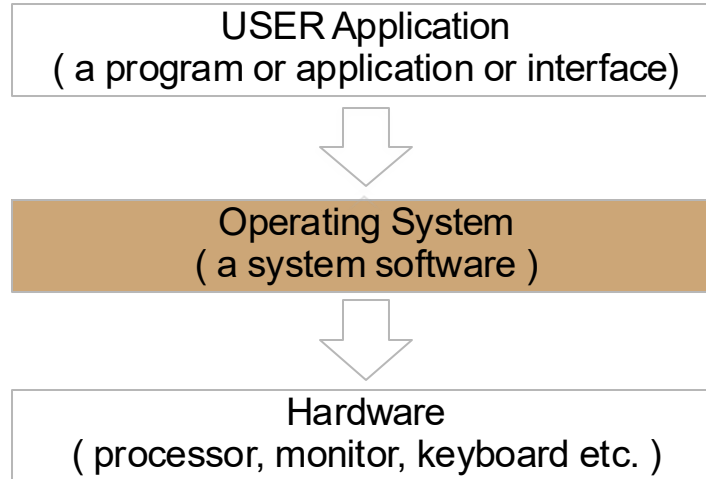
CSE 323

Operating Systems

Lecture 2: Introduction to OS

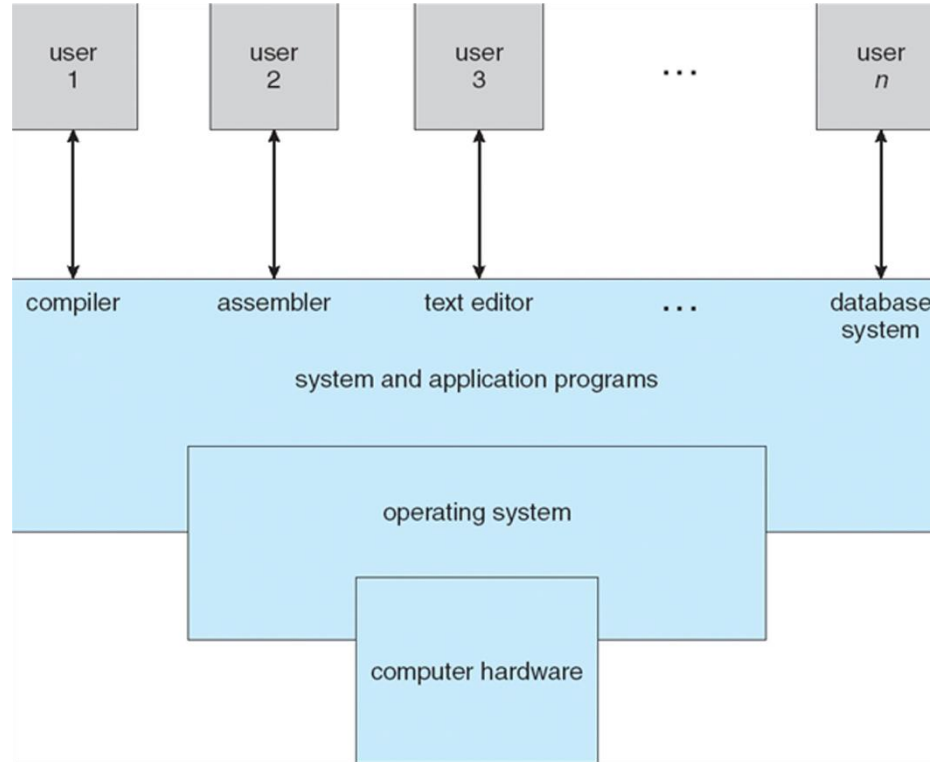
What is an Operating System?

A program that acts as an intermediary
between a user of a computer and the computer hardware



Computer System Organization

Components of a Computer System



Kernel

The one program running at all times.

- Kernel is the central module of an operating system.
- Part of OS that loads first, and it remains in main memory
- As small as possible
- Provide all the essential services required by other parts of the operating system and applications
- Kernel code is usually loaded into a protected area of memory to prevent it from being overwritten

Bootstrap Program

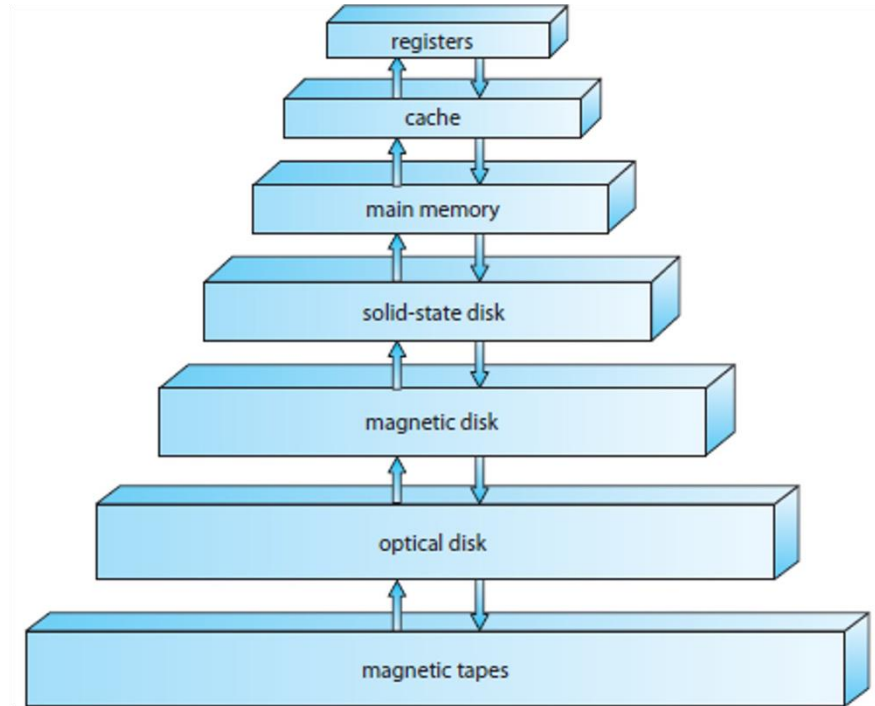
An initial program executed when a computer starts running

- When a computer is powered up or rebooted, it is executed first
- Stored in the ROM or EEPROM, known as firmware
- Initializes all aspects of the system, from CPU registers to device controllers to memory contents
- bootstrap program must know how to load the operating system and how to start executing
- Once the OS kernel is loaded and executing, it can start providing services to the system and its users

Storage Structure

- ❑ Main memory – only large storage media that the CPU can access directly
 - ❑ Random access
 - ❑ Typically volatile
 - ❑ Secondary storage – extension of main memory that provides large nonvolatile storage capacity
-
- CPU can load instructions only from main memory
 - General-purpose computers run most of their programs from rewritable memory, called main memory (also called RAM)
 - Computers use other forms of memory as well - Read Only Memory (ROM) and electrically erasable programmable read-only memory (EEPROM)
 - Only static programs, such as the bootstrap program described earlier, are stored here

Storage Device Hierarchy

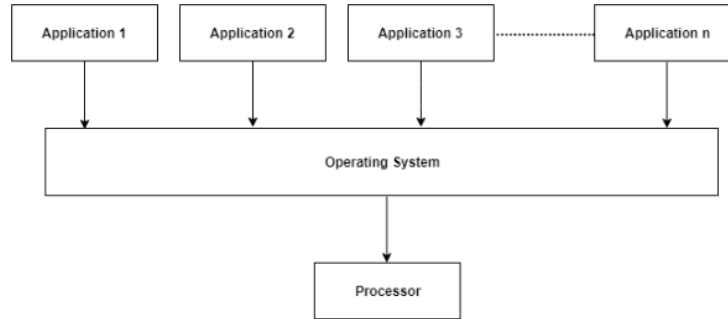


Operating Systems
OS Architecture

Operating System Architecture

Single-Processor Systems:

- One main CPU capable of executing a general-purpose instruction set.
- Almost all single processor systems have other special-purpose processors (device-specific processors), which run a limited instruction set.

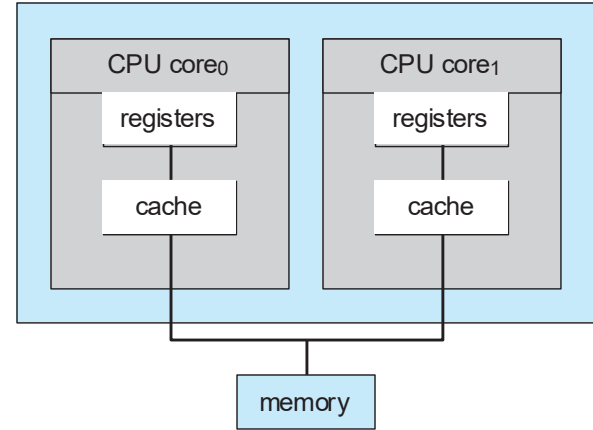
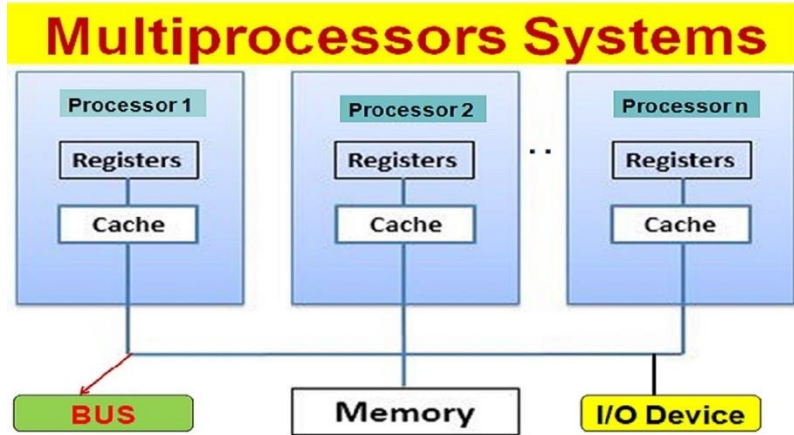


Single Processor System

Operating System Architecture

Multiprocessor Systems (parallel systems or multicore systems):

- Such systems have **two or more processors in close communication**, sharing the computer bus and sometimes the clock, memory, and peripheral devices.

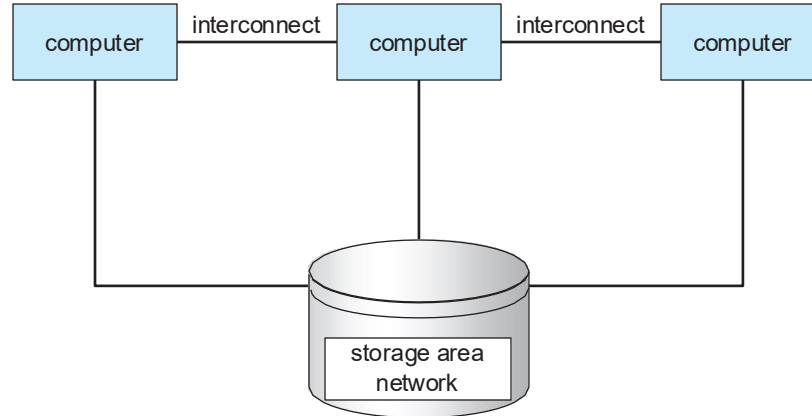


A dual-core system

Operating System Architecture

Clustered Systems:

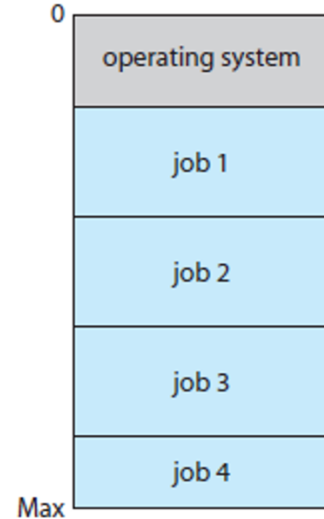
- Special kind of multiprocessor system which **gathers together multiple CPUs**. They are composed of two or more individual systems- or nodes - joined together.
- Clustered computers share storage and are closely linked via a local-area network (LAN) or a faster interconnect, such as InfiniBand



Operating System Structure

Multiprogramming:

- Increases CPU utilization by organizing jobs (code and data) so that the CPU always has one to execute
- OS keeps **several jobs in memory simultaneously**. As main memory is small, it keeps the jobs on the disk (job pool) which waits there to be allocated in the main memory
- OS picks and begins to execute one of the jobs in memory. Eventually, the job may have to wait for some task. OS picks another job from the pool to execute while the previous one waits.



Memory layout for a multiprogramming system.

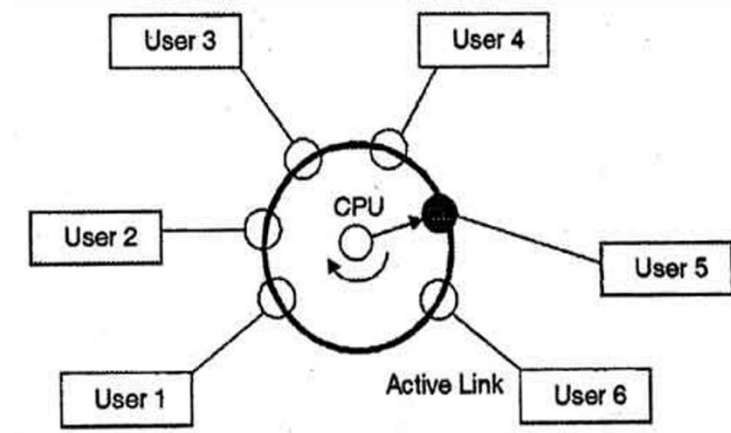
Requirements of Multiprogramming

- **Job Scheduling:** If several jobs are ready to be brought into memory, and if there is not enough room for all of them, then the system must choose among them.
 - When OS selects a job from the job pool, it loads that job into memory for execution. Having several programs in memory at the same time requires some form of memory management.
 - **CPU Scheduling:** if several jobs are ready to run at the same time, the system must choose which job will run first.
 - Running multiple jobs concurrently requires that their ability to affect one another be limited in all phases of the operating system.
- ★ If processes don't fit in memory, swapping moves them in and out to run from main memory achieving this goal is virtual memory.

Operating System Structure

Time Sharing:

- CPU executes multiple jobs by switching among them.
- Switches occur so frequently that the users can interact with each program while it is running.
- requires an interactive computer system, which provides direct communication between the user and the system.
- Response time should be short.

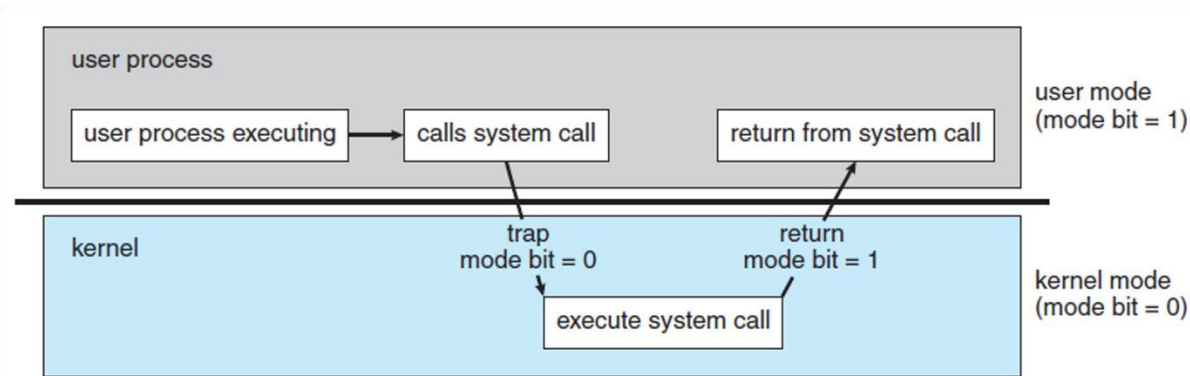


Operating System Operations

- Modern operating systems are interrupt driven
- Events are almost always signaled by the occurrence of an interrupt or a trap
- A trap (or an exception) is a software-generated interrupt
- For each type of interrupt, separate segments of code determine what action should be taken
- Errors can occur when one erroneous program modify another program, the data of another program, or even the operating system itself
- A properly designed operating system must ensure that an incorrect (or malicious) program cannot cause other programs to execute incorrectly

Dual Mode Operation

- Need to distinguish between the execution of operating-system code and user defined code
- A bit, called the mode bit, is added to the hardware of the computer (i. e. CPU) to indicate the current mode: kernel (0) or user (1)
- dual mode of operation provides protection of the operating system from errant users
- this protection is provided by designating some of the machine instructions that may cause harm as privileged instructions that are executed only in kernel mode

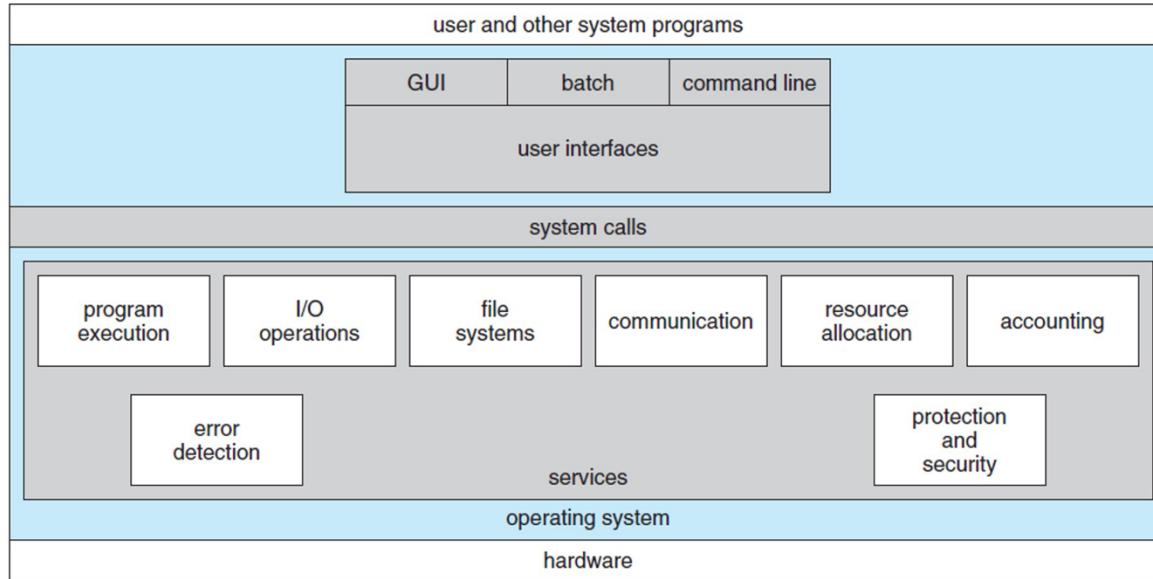


Transition from user to kernel mode.

OS Services

Operating System Services

- OS provides an environment for the execution of programs.
- Specific services provided, differ from one operating system to another, but there are some common classes
- Services are provided for the convenience of the programmer

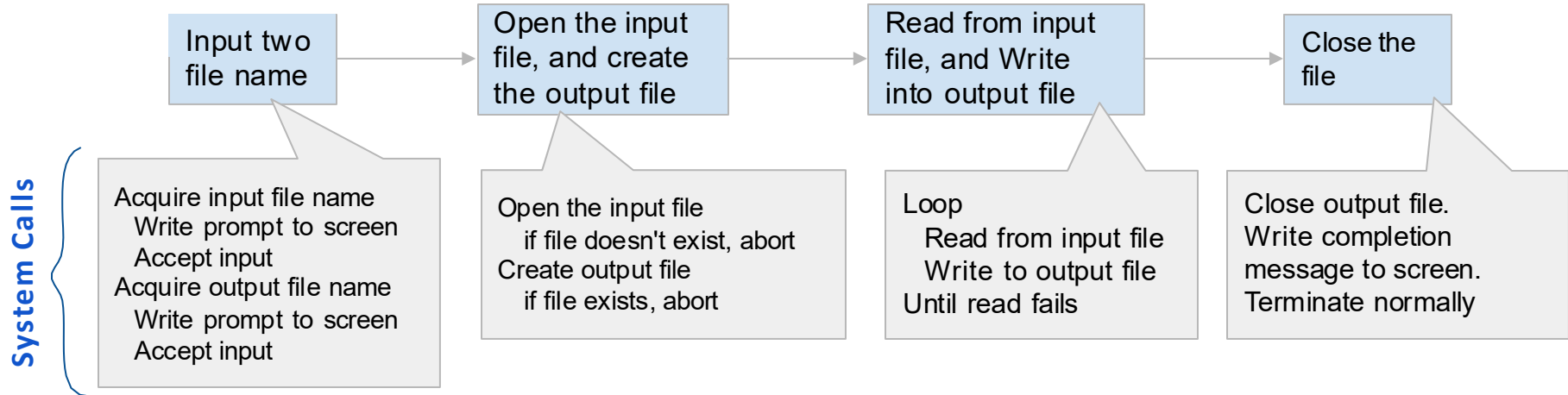


System Call, System Program, System Boot

System Call

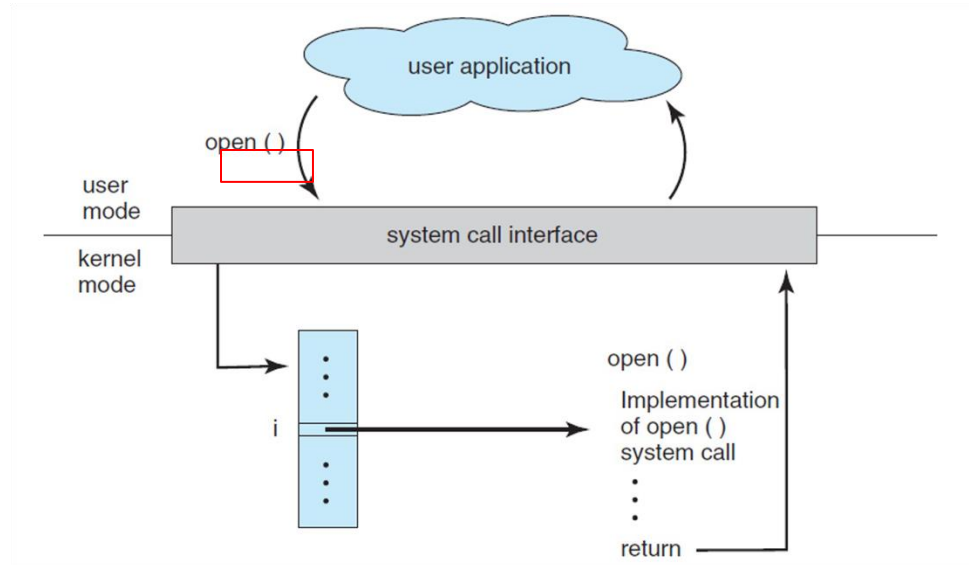
- System calls provide an **interface** to the services made available by an operating system
- These calls are generally available as routines/functions
- Routines are written in C or C++. Some low level tasks are written in assembly language

A program to copy the contents of a file to another file !



System Call Interface

- Serves as the link to system calls made available by the operating system
- A number is associated with each system call, and the system-call interface maintains a table indexed according to these numbers
- Invokes the intended system call in the operating-system kernel and returns the status of the system call and any return values



Types of System Call

Type	Windows OS	Linux OS
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()

Types of System Call

Type	Windows OS	Linux OS
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shm_open() mmap()
Protection	SetFileSecurity() InitializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()

System Programs

System programs, also known as system utilities, provide a convenient environment for program development and execution.

These system programs provide -

- File management
- Status information
- File modification
- Programming-language support
- Program loading and execution
- Communications
- Background services

System Boot

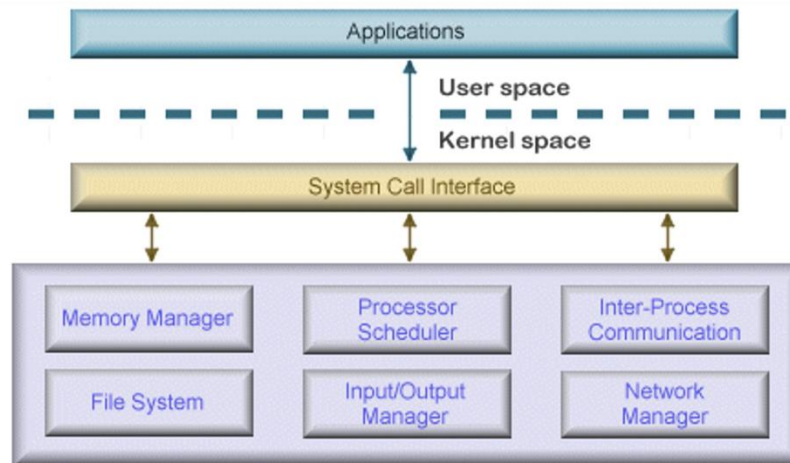
- When power initialized on system, execution starts at a fixed memory location.
 - Firmware ROM used to hold initial boot code
- Operating system must be made available to hardware so hardware can start it.
 - Small piece of code – **bootstrap loader**, stored in ROM locates the kernel, loads it into memory, and starts it
 - Sometimes two-step process where **boot block** at fixed location loaded by ROM code, which loads bootstrap loader from disk
- Common bootstrap loader, **GRUB**, allows selection of kernel from multiple disks, versions, kernel options
- Kernel loads and system is then running.

OS Structures

OS Structure

Simple/Monolithic structure:

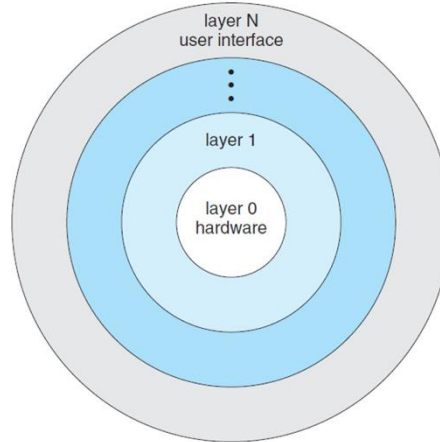
- Earliest and most common architecture
- Every component of OS is in the kernel and can communicate with each other directly
- Complex and Large (millions line of code ,) hard to maintain



OS Structure

Layered structure:

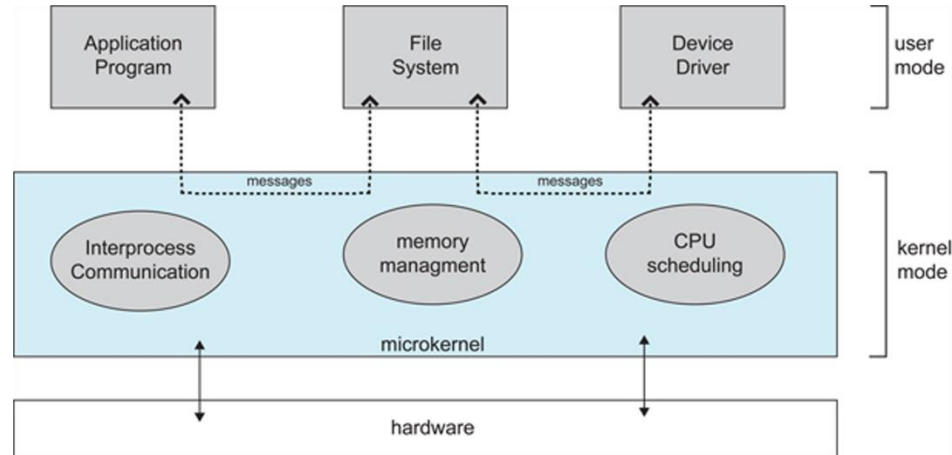
- OS is divided into layers.
- Each layer can use services of its lower layers.
- Easy to debug and develop.
- Less efficient as each layer adds some overhead



OS Structure

Microkernel structure:

- Moves as much from kernel into user space
- Communication takes place between user modules using message passing
- Easier to extend a microkernel
- More reliable(less code running in kernel mode) and secure
- Performance overhead



Questions?